

THE FLIP MISSION TO MONS MOUTON: PROJECT UPDATE. Luke Walker¹, Kelly Randell², Manny Shar³.
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Introduction: The FLEX Innovation Platform (FLIP) Mission is a unique mission of opportunity to the Mons Mouton region of the lunar south pole scheduled to launch in mid-2026. Venturi Astrolab Inc. (Astrolab) conceived of the mission in response to the VIPER cancellation in 2024 and has designed, built, and prepared the vehicle for protoqualification testing in under 20 months. FLIP will serve several critical functions for the future of lunar south pole exploration: 1) science investigations for lunar environments investigations, 2) technology maturation of key technologies for future lunar rovers including NASA's Lunar Terrain Vehicle (LTV), and 3) characterizing performance of mobility, perception, and thermal systems in the challenging lunar south pole environment. In the process, Astrolab is gaining valuable corporate experience in designing, building, qualifying, and operating a lunar mobility vehicle.

We provide an update on the FLIP mission, including: current vehicle build and test status, the payloads and science investigations on FLIP, and the operational mission concept.

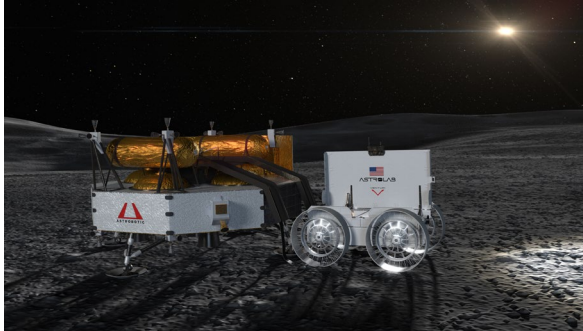


Figure 1 FLIP drives away from the Astrobotic Griffin lander

Vehicle Build and Test: The FLIP rover is a unique design informed by the constraints in its conception and purpose as a technology development platform. NASA cancelled VIPER in 2024 due to cost increases, launch delays, and risks of future cost growth [1], but maintained the Astrobotic Griffin Lander as a part of the Commercial Lunar Payload Services (CLPS) program. Astrolab contracted with Astrobotic for the available space on Griffin and began designing a vehicle.

FLIP is a four-wheeled skid steer rover limited to the geometry and launch mass (480kg) initially designed for VIPER. FLIP includes mobility components designed and sized for the larger FLEX vehicle, including

innovative warm-blooded flexible tires from Astrolab's strategic partner Venturi Space and high-power wheel actuator. The remaining systems were selected for applicability to future vehicles and ability to procure within the limited timeframe of development.

The vehicle is currently going through Systems Assembly, Integration and Test (AI&T), including payload integration and testing. Current results from the AI&T campaign will be presented as well as future plans for integrated testing prior to launch.



Figure 2 The FLIP rover in integration at Astrolab.

Payloads and Science Investigations: FLIP hosts thirteen payloads, including science instruments associated with four separate NASA centers, as well as commercial and outreach opportunities. FLIP will carry out important investigations for dust, perception, lunar environments, and regolith composition. The science instruments and experiments will be discussed in more detail.

Operational Concept: As a mission of opportunity, Astrolab is developing FLIP's operational concept for flexibility in landing times, communications availability, and lunar hibernation. Payload operations and science investigations will be integrated with lunar traverses. The overall mission plan, including mission phases and timelines, traverse plans, and integrated payload operations will be discussed.

References:

- [1] NASA, 2024, "NASA Ends VIPER Project, Continues Moon Exploration", <https://www.nasa.gov/news-release/nasa-ends-viper-project-continues-moon-exploration/>